

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MENTALCLOUDS | **Context:** Global South Mental Health Practitioners — India-Anchored OMHC / Peer-Support Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output ontology mirrors the input ontology (three reference summaries per session aligned to the same three counseling components [Q15, Q16]) and inherits its structural gaps — risk-flag content, family-intervention suggestions, and spiritual/religious coping have no scoring representation. The authors confirm models 'are not suitable for clinical use as they stand now' due to poor scores on overall efficacy and opportunity cost [Q51], and note suicide-risk and substance-use content was not recorded [Q64]. The Sekhon et al. clinical acceptability framework [Q44, Q48] adds general dimensions (affective attitude, burden, ethicality, coherence, opportunity costs, perceived effectiveness) and a hallucination dimension [Q46, Q53], but none of these directly score risk-flag recall, family-intervention completeness, or cultural-content fidelity. The structure-separation problem [Q62] also undermines the scoring schema's coherence. Empirical evidence shows LLMs systematically over-rate appropriateness of responses to suicidal content vs. expert suicidologists [WEB-13], compounding the scoring gap. Given OO is HIGH priority and the gap is structural, the dimension is significantly misaligned.

ID	Evidence
Q15	"Within each conversation, three pivotal counseling components (aspects) emerge – symptom and history exploration, patient discovery, and reflective utterances." (p.6)
Q16	"Our study aims to capture the essence of each aforementioned counseling component, embarking on the creation of three distinct summaries for a single dialogue — each tailored to a specific counseling component." (p.6)
Q44	"The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)
Q46	"Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed)." (p.12)
Q48	"The clinical acceptability framework [39] involves six parameters – affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness (c.f. Table 5 for more details)." (p.13)
Q51	"As all three models have poor scores on the more sensitive aspects i.e. the overall efficacy and the opportunity cost, this indicates that these models share the same weakness and are not suitable for clinical use as they stand now." (p.14)
Q53	"Additionally, the evaluation of hallucination identification is divided into three categories: no hallucination observed, hallucination barely observed, and too much hallucinated in a set of 39 conversations." (p.14)
Q62	"However, the models did not do as well with the structure separation of the information. The sections of "symptoms and history", "patient discovery", and "reflection" frequently overlap, posing clinical and legal problems." (p.15)
Q64	"Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified." (p.15)
WEB-13	LLMs systematically over-rate appropriateness of responses to suicidal ideation vs. expert suicidologists (https://www.jmir.org/2025/1/e67891)

Strengths

- The dual quantitative-qualitative scoring design [Q5, Q18] is a methodological strength: ROUGE/BERTScore are supplemented by a six-parameter clinical acceptability framework [Q44, Q48] and a hallucination dimension [Q46], providing more multidimensional assessment than typical summarization benchmarks.

- Per-component scoring [Q26, Q31, Q40] enables granular diagnosis of which counseling aspects models handle well, supporting partial fitness diagnosis even with the taxonomy gap.
- Hallucination categorization [Q53, Q54] partially addresses the patient-safety dimension that pure ROUGE would miss.

ID	Evidence
Q5	"The generated summaries are evaluated quantitatively using standard summarization metrics and verified qualitatively by mental health professionals." (p.1)
Q18	"We undertake a comprehensive evaluation of the generated summaries across various architectures, employing a dual approach of quantitative and qualitative assessments." (p.9)
Q26	"Table 2 reports the automatic evaluation scores of LLMs on the summarization task for the Symptom and History (SH) psychotherapy element." (p.10)
Q31	"The experimental results presented in Table 3 focus on the summarization task for the Patient Discovery (PD) psychotherapy element." (p.10)
Q40	"Table 4. Results obtained on MentalCLOUDS for the summarization task on the Reflecting (RT) psychotherapy element." (p.11)
Q44	"The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)
Q46	"Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed)." (p.12)
Q48	"The clinical acceptability framework [39] involves six parameters – affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness (c.f. Table 5 for more details)." (p.13)
Q53	"Additionally, the evaluation of hallucination identification is divided into three categories: no hallucination observed, hallucination barely observed, and too much hallucinated in a set of 39 conversations." (p.14)
Q54	"These categories essentially determine how well the response is consistent with the context and whether it is also incorrect, nonsensical, or contains global information beyond the scope of the conversation." (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The three counseling-component output categories [Q15] are partially regionally relevant (CBT-anchored, recognized by primary cohort per A3) but omit categories that the deployment requires.

OO-2: Missing categories for regional deployment: risk-flag documentation [Q64], family intervention notes (structural absence), spiritual/religious coping (structural absence given American-sourced inputs [Q70, WEB-15]), therapy-technique identification [Q63]. National policy context (MHCA 2017 Section 115 [WEB-10]; NSPS 2022 [WEB-30]) elevates the severity of the risk-flag omission.

OO-3: The clinical acceptability framework's six parameters [Q48] are general and not obviously encoding non-regional values, but they were not adapted to scoring South Asian family-systems or faith-coping content; the absence of such adaptation implicitly defaults to a Western/individualist clinical framing.

OO-4: Stakeholder-driven taxonomy redesign is warranted: the elicitation explicitly identifies risk, family, and spiritual categories as required (A2), and the benchmark's own authors acknowledge the gap [Q64, Q69].

OO-5: Documented harms to structural and content validity: scoring schema provides no signal on safety-critical content [Q64]; structural overlap among the three categories [Q62]; expert-confirmed clinical unfitness 'as they stand now' [Q51]; empirical

evidence that LLMs miscalibrate suicidal-content responses [WEB-13].

ID	Evidence
Q15	"Within each conversation, three pivotal counseling components (aspects) emerge – symptom and history exploration, patient discovery, and reflective utterances." (p.6)
Q44	"The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)
Q48	"The clinical acceptability framework [39] involves six parameters – affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness (c.f. Table 5 for more details)." (p.13)
Q51	"As all three models have poor scores on the more sensitive aspects i.e. the overall efficacy and the opportunity cost, this indicates that these models share the same weakness and are not suitable for clinical use as they stand now." (p.14)
Q62	"However, the models did not do as well with the structure separation of the information. The sections of "symptoms and history", "patient discovery", and "reflection" frequently overlap, posing clinical and legal problems." (p.15)
Q63	"The models are also unable to identify psychotherapy types (e.g., cognitive behavior therapy) and therapy techniques, which form an integral part of counseling notes." (p.15)
Q64	"Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified." (p.15)
Q69	"Finally, this work explored only three aspects (counseling component) of the conversation." (p.16)
Q70	"Additionally, the counseling sessions in this work represented a certain demographic region (American) and thus may not apply to therapy counseling for other demographics." (p.16)
WEB-10	MHCA 2017 Section 115 imposes duty-of-care on suicide risk (https://pmc.ncbi.nlm.nih.gov/articles/PMC6482674/)
WEB-13	LLMs systematically over-rate appropriateness of responses to suicidal ideation vs. expert suicidologists (https://www.jmir.org/2025/1/e67891)
WEB-15	Faith-healer integration documented in Indian mental health practice (https://pmc.ncbi.nlm.nih.gov/articles/PMC7616029/)
WEB-30	India National Suicide Prevention Strategy 2022 (https://www.pib.gov.in/PressReleasePage.aspx?PRID=2188003)

Information Gaps

- Whether the Sekhon et al. parameter ratings implicitly captured culturally salient content quality (e.g., did 'perceived effectiveness' downgrade summaries that omitted family/spiritual content?) — not documented.

Requires Expert Verification

- Indian clinical experts' view on whether the six-parameter framework can be retrofitted with risk/family/spiritual sub-criteria, or whether a redesigned taxonomy is required.

Remediation

Gap 1: Scoring frame provides no signal on risk-flag recall, family-intervention completeness, or spiritual-coping coverage; existing components frequently overlap.

Recommendation: Add discrete scoring dimensions for safety-critical content recall, family-intervention completeness, and cultural-coping coverage. Define explicit decision rules to disambiguate overlapping categories before re-scoring.